functional diagram.

→ The inputs $(x=y)_{in}$, $(x>y)_{in}$ and $(x<y)_{in}$ are used when we need to connect more than one IC 7485 to compare numbers having more than 4-bits.

☆ Applications:

→ Comparators are used in central processing units (CPUs) & microcontrollers (MCUs).

→ These are used in control applications in which the binary numbers representing physical variables such as temperature, position etc, are compared with a reference value.

→ Used in password verification & biometric applications.

→ Servo-motor control

→ It is a major element in ALU.

→ Used for sorting.

→ Used in the field of parallel processing & multiprocessing.

Unit - 2

Mux Tree:-

→ Mux tree is nothing but obtaining higher order mux using lower order mux.

Problems:-

1) Implement 4x1 mux using 2x1 mux.

Required mux : 4:1 mux

Available mux : 2:1 mux

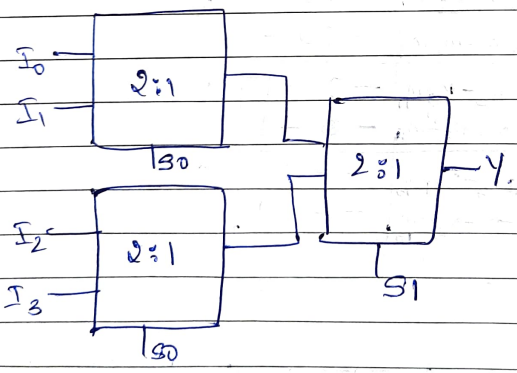
4:1 MUX TT

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$n = 4$

$4/2 = 2$

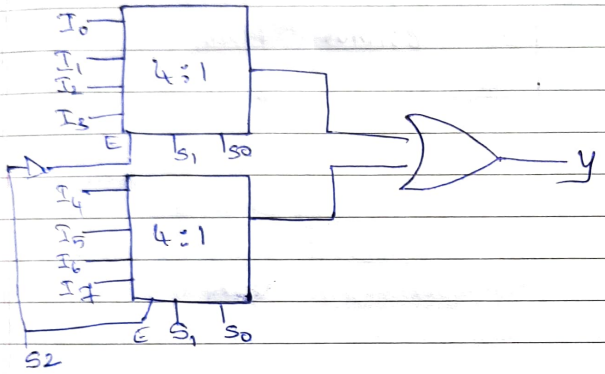
$2/2 = 1$ } $2+1=3$



② Implement 8:1 MUX using 4:1 MUX

Required MUX: 8:1 MUX
Available MUX: 4:1 MUX

n = 8 = 8/4 = 2
2/4 = 0.5
2 + 0.5 = 2.5

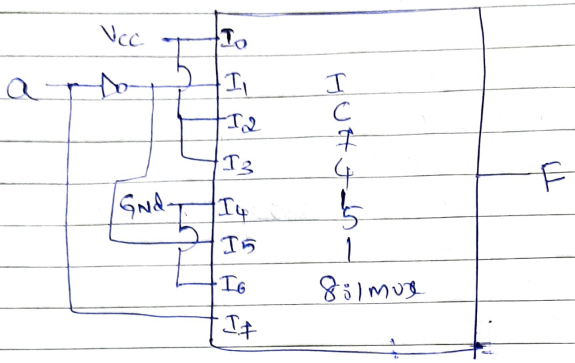


③ Implement the following boolean equation using 8:1 multiplexer.

$Y = F(a,b,c,d) = \Sigma m(1,2,3,5,10,11,15)$

Here a is input, b, c & d are data select lines.

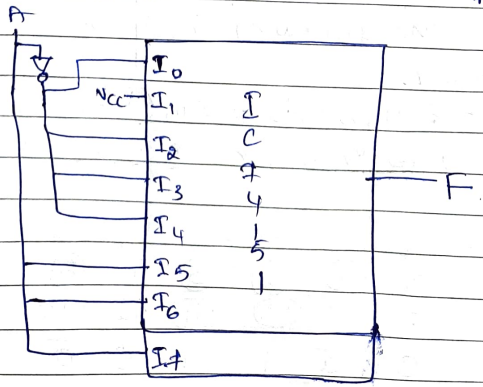
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{a}$	0	(1)	(2)	(3)	4	(5)	6	7
a	8	9	(10)	(11)	12	13	14	(15)
	0	$\bar{a}$	1	1	0	$\bar{a}$	0	a



④ Implement the following boolean equation using 8:1 multiplexer - $Y = F(A,B,C,D) = \Sigma m(0,1,2,3,4,9,13,14,15)$

Here A is input, B, C & D are data select lines.

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{A}$	(0)	(1)	(2)	(3)	(4)	5	6	7
A	8	(9)	10	11	12	(13)	(14)	(15)
	$\bar{A}$	1	$\bar{A}$	$\bar{A}$	$\bar{A}$	A	A	A



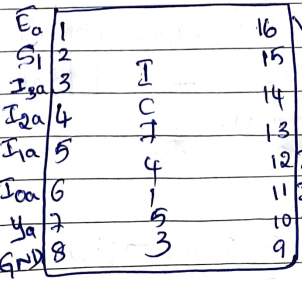
★ Difference between Multiplexer and Demultiplexer

Parameter	Multiplexer	Demultiplexer.
Definition	Multiplexer refers to a type of combinational circuit that accepts multiple inputs of data but provides only a single output.	It accepts just a single input but directs it through multiple outputs.
Common name	Data Selector	Data distributor
Operational Principle	It works on an operational principle of many-to-one	It works on an Operational principle of one to many
No of Data inputs	It has multiple inputs of data and signals	It has a single input of data & signals.
No of Data outputs	It generates a single output for data and signals	It generates multiple outputs for data & signals
Information processing	It processes the digital data & info by collecting them from multiple sources & integrating them into a single source as the output	It collects digital data & info from one single source/channel & then converts it into a set of multiple sources as the outputs.
Type of Digital Setup	The multiplexer acts as a digital switch	The demultiplexer acts as a digital circuit.
End of usage	In the process of time-division multiplexing, we use a multiplexer at the end of the transmitter.	In the process of time-division multiplexing, we use a demultiplexer at the end of the receiver.

★ Difference between Demultiplexer and Decoder.

Parameter	Decoder	Demultiplexer.
Basic	These are logic circuit which decodes an encrypted input stream from one to another format.	It is a combination circuit which routes a single input signal to one of several output signals
Input / Output	n number of input lines and 2^n number of output lines.	n number of select lines and 2^n number of output lines.
Inverse of	Encoder	Multiplexer.
Application	In detection of bits, data encoding.	In distribution of the data, Switching.
Use	It is used for changing the format of the instruction in the machine specific language.	It is used as a routing device to route the data coming from one signal into multiple signals.
Select lines	Do not contains	It contains
Implementation	Majorly implemented in the networking application	Employed in data-intensive applications where data need to be changed into another form.

(5) Design & implement a full adder and a full subtractor using multiplexer (Dual 4:1 mux - IC 74153)



$\bar{E}_a, \bar{E}_b$ = Active low enable i/p

S_0, S_1 - Select Input

Y_a, Y_b = Outputs

$I_{0a} - I_{3a}$ & $I_{0b} - I_{3b}$ - Inputs.

i) Full adder Truth Table:

A	B	C	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$S(A,B,C) = \sum m(1,2,4,7)$

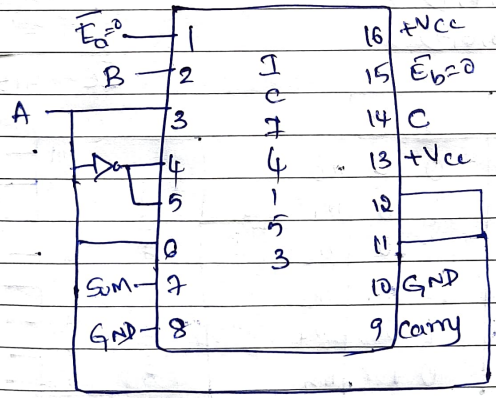
$C_{out}(A,B,C) = \sum m(3,5,6,7)$

Sum(S)

	I_{0a}	I_{1a}	I_{2a}	I_{3a}
$\bar{A}$	0	(1)	(2)	3
A	(4)	5	6	(7)
	A	$\bar{A}$	$\bar{A}$	A

Carry(C_{out})

	I_{0b}	I_{1b}	I_{2b}	I_{3b}
$\bar{A}$	0	1	2	(3)
A	4	(5)	(6)	(7)
	0	A	A	1



→ Here B, C are selection inputs
→ A is input.

ii)

A	B	C	B	D
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Full Subtractor.

$B(A,B,C) = \sum m(1,2,3,7)$

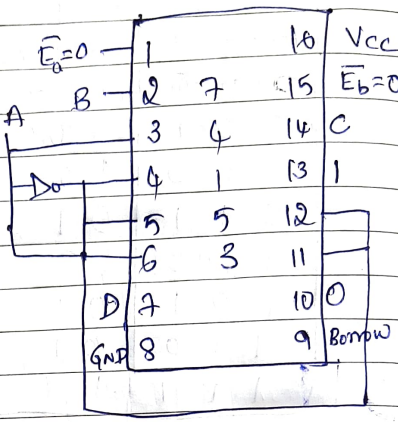
$D(A,B,C) = \sum m(1,2,4,7)$

Difference D

	I_{0a}	I_{1a}	I_{2a}	I_{3a}
$\bar{A}$	0	(1)	(2)	3
A	(4)	5	6	(7)
	A	$\bar{A}$	$\bar{A}$	A

Borrow B.

	I_{0b}	I_{1b}	I_{2b}	I_{3b}
$\bar{A}$	0	(1)	(2)	(3)
A	4	5	6	(7)
	0	$\bar{A}$	$\bar{A}$	1

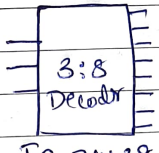


6) Implement Full adder and Full Subtractor using decoder

1) Full adder

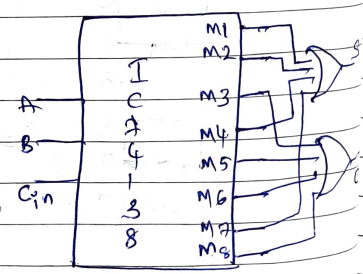
$S = \Sigma m(1,2,4,7)$

$C = \Sigma m(3,5,6,7)$

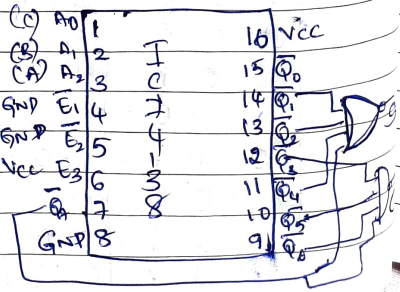


IC-74138

A	B	C_{in}	S	C	
0	0	0	0	0	M_0
0	0	1	1	0	M_1
0	1	0	1	0	M_2
0	1	1	0	1	M_3
1	0	0	1	0	M_4
1	0	1	0	1	M_5
1	1	0	0	1	M_6
1	1	1	1	1	M_7



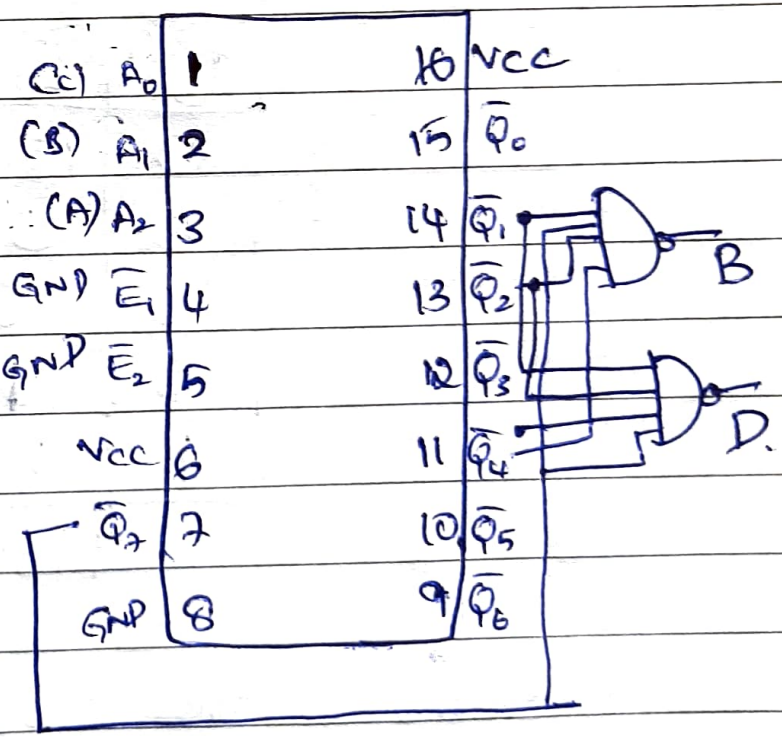
Note:
All are active low signals except E_3.



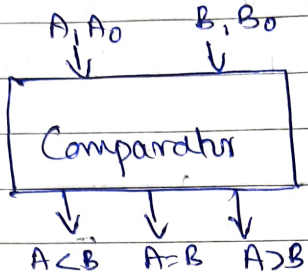
ii) Full subtractor using 3:8 decoder

A	B	C	B	D
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$F(A,B,C) = \Sigma m(1,2,3,7)$
$\Sigma m(1,2,4,7)$



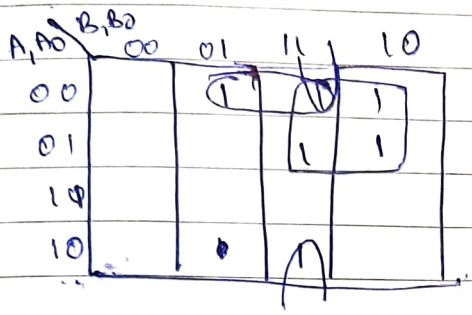
☆ Two-Bit Magnitude Comparator.



A_1	A_0	B_1	B_0	A<B	A=B	A>B
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

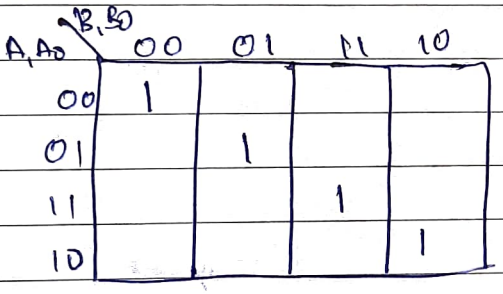
Boolean equations using K-Map

i) $A < B$



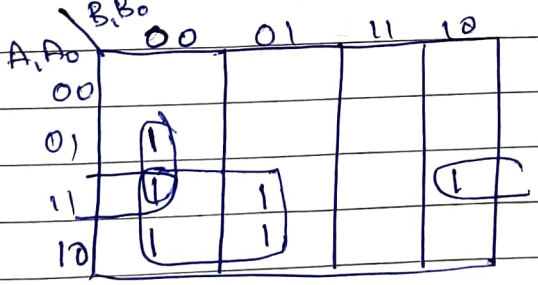
$A<B = \bar{A}_1\bar{A}_0B_0 + \bar{A}_1B_1 + \bar{A}_0B_1B_0$

ii) $A = B$



$A = B = \overline{(A_1 \oplus B_1)}\,\overline{(A_0 \oplus B_0)}$

iii) $A > B$



$A>B = A_0\bar{B}_1B_0 + A_1A_0\bar{B}_0 + A_1\bar{B}_1$